Publications

Journal Articles

[1] **Tomas Monopoli**, Xinglong Wu, Cheng Yang, Christian Schuster, Sergio Amedeo Pignari, Johannes Wolf, and Flavia Grassi. "Identification of Critical Frequencies in Wideband Near-Field Spatial Scans". In: *IEEE Transactions on Electromagnetic Compatibility* 68.3 (2026), pp. 652–663. `doi:10.1109/TEMC.2026.3673588`.

[2] **Tomas Monopoli**, Xinglong Wu, Sergio Amedeo Pignari, Johannes Wolf, and Flavia Grassi. "Efficient Electromagnetic Near-Field Scanning Using Physics-Informed Gaussian Process Regression". In: *IEEE Access* 13 (2025), pp. 191781–191794. `doi:10.1109/ACCESS.2025.3629347`.

[3] Cheng Yang, **Tomas Monopoli**, Sebastian Götschel, Xinglong Wu, Flavia Grassi, and Christian Schuster. "Adaptive On-the-Fly Scan Method for Fast and Efficient Planar Near-Field Acquisition". In: *IEEE Antennas and Wireless Propagation Letters* 24.8 (2025), pp. 2143–2147. `doi:10.1109/LAWP.2025.3557192`.

[4] **Tomas Monopoli**, Xinglong Wu, Cheng Yang, Christian Schuster, Sergio Amedeo Pignari, Johannes Wolf, and Flavia Grassi. "Morphological Search for Near-Field Equivalent Infinitesimal Dipole Models". In: *IEEE Transactions on Electromagnetic Compatibility* (2024), pp. 1–12. ISSN: 1558-187X. `doi:10.1109/TEMC.2024.3492701`.

[5] **Tomas Monopoli**, Xinglong Wu, Flavia Grassi, Sergio Amedeo Pignari, and Johannes Wolf. "A New Method Exploiting Partial Image Expansion to Include Substrate and Ground in Dipole-Based Near-Field Models". In: *IEEE Transactions on Electromagnetic Compatibility* 65.6 (2023), pp. 1878–1887. `doi:10.1109/TEMC.2023.3325242`.

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[6] **Tomas Monopoli**, Jiahong Liu, and Shuai Zhao. "A Bayesian Gradient-Based Framework for Probabilistic Parameter Identification of DC–DC Converters". In: *2026 IEEE Energy Conversion Congress and Exposition (ECCE).* Accepted for publication. Vancouver, BC, Canada, Oct. 2026.

[7] **Tomas Monopoli**, Xinglong Wu, Sergio A. Pignari, Johannes Wolf, and Flavia Grassi. "Wideband Modeling and Simulation of Multiple PCB Components in CubeSat Environment". In: *2025 ESA Workshop on Aerospace EMC (Aerospace EMC).* 2025, pp. 1–6. `doi:10.23919/AerospaceEMC64918.2025.11074834`.

[8] **Tomas Monopoli**, Xinglong Wu, Sergio Pignari, Johannes Wolf, and Flavia Grassi. "Inspection tools for Gaussian Process Regression Modeling of Electromagnetic Fields of Electronic Boards and Chips". In: *2024 14th International Workshop on the Electromagnetic Compatibility of Integrated Circuits (EMC Compo).* Oct. 2024, pp. 56–59. `doi:10.1109/EMCCompo61192.2024.10742047`.

[9] **Tomas Monopoli**, Xinglong Wu, Cheng Yang, Sergio A. Pignari, Johannes Wolf, and Flavia Grassi. "Partial Image Expansion Applied to Real Near-Field Measurement Data Modeling". In: *2024 IEEE Joint International Symposium on Electromagnetic Compatibility, Signal & Power Integrity: EMC Japan / Asia-Pacific International Symposium on Electromagnetic Compatibility (EMC Japan/APEMC Okinawa).* 2024, pp. 138–141. `doi:10.23919/EMCJapan/APEMCOkinaw58965.2024.10585137`.

[10] **Tomas Monopoli**, Xinglong Wu, Flavia Grassi, Sergio A. Pignari, and Karl-Friedrich Johannes Wolf. "Positioning Uncertainty of Near-Field Probes". In: *2023 IEEE 7th Global Electromagnetic Compatibility Conference (GEMCCON).* 2023, pp. 57–57. `doi:10.1109/GEMCCON57842.2023.10078196`.

[11] **Tomas Monopoli**, Antonino Borgese, Xiaokang Liu, Flavia Grassi, and Sergio A. Pignari. "Modal Analysis of a Typical Power over Ethernet Configuration". In: *2020 6th Global Electromagnetic Compatibility Conference (GEMCCON).* 2020, pp. 1–4. `doi:10.1109/GEMCCON50979.2020.9456742`.